

AUGHTON, United Kingdom

WMO: 033220

Lat: 53.550N

Lon: 2.917W

Elev: 184

StdP: 14.60

Time Zone: 0.00 (EUW)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.8	29.3	20.5	15.5	29.4	23.3	17.7	31.5	32.5	44.8	28.5	44.4	7.7	120	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	76.0	63.3	72.3	62.1	68.9	60.6	64.9	73.3	63.4	70.2	61.8	67.4	9.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	83.8	67.0	60.5	79.2	65.7	59.2	75.5	64.6	29.9	73.7	28.8	70.1	27.7	67.5	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.7	22.9	20.4	DB	23.5	81.5	4.4	4.7	20.3	84.9	17.8	87.7	15.3	90.3	12.1	93.7	(4)
(5)			WB	22.4	68.0	4.2	1.4	19.4	69.0	17.0	69.9	14.6	70.7	11.6	71.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	40.3	39.9	43.3	46.3	52.5	56.5	60.6	59.6	55.2	50.6	44.9	41.6	(6)
(7)		DBStd	8.74	5.54	5.72	4.51	4.96	5.18	4.98	4.22	4.61	3.86	4.79	5.13	5.72	(7)
(8)		HDD50	1443	300	283	211	133	29	3	0	0	4	51	164	265	(8)
(9)		HDD65	5748	765	702	672	560	387	262	150	182	294	445	603	726	(9)
(10)		CDD50	1202	1	1	4	23	107	197	327	297	160	70	11	4	(10)
(11)		CDD65	33	0	0	0	0	0	6	13	14	0	0	0	0	(11)
(12)		CDH74	217	0	0	0	0	5	41	65	102	4	0	0	0	(12)
(13)		CDH80	35	0	0	0	0	0	3	6	26	0	0	0	0	(13)
(14)	Wind	WSAvg	10.0	12.2	11.2	12.0	10.3	9.3	8.6	8.9	8.6	9.0	9.1	9.9	10.7	(14)
(15)	Precipitation	PrecAvg	42.1	4.1	3.1	2.6	2.4	2.7	2.9	3.2	3.8	3.7	4.5	4.4	4.7	(15)
(16)		PrecMax	55.7	7.9	7.0	6.0	4.1	4.6	6.8	7.0	8.0	7.7	10.3	8.9	10.6	(16)
(17)		PrecMin	32.5	0.6	0.5	0.9	0.8	0.6	1.1	1.1	1.0	0.6	1.5	2.2	1.3	(17)
(18)		PrecStd	6.4	1.7	1.5	1.3	0.9	1.1	1.5	1.3	1.4	1.5	1.9	1.8	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	55.0	58.6	68.1	74.2	79.5	80.4	83.7	72.8	65.0	57.6	56.8	(19)
(20)			MCWB	50.6	51.2	52.1	57.9	61.6	63.6	65.6	65.6	63.2	59.0	54.3	53.1	(20)
(21)		2%	DB	51.9	52.3	53.7	61.8	69.8	73.6	75.2	75.5	67.0	61.7	55.8	54.2	(21)
(22)			MCWB	48.8	48.6	49.4	53.9	58.9	62.8	63.6	63.2	59.8	57.0	53.0	51.3	(22)
(23)		5%	DB	50.0	50.5	52.0	57.2	64.5	68.5	71.7	70.4	63.7	59.5	53.8	51.7	(23)
(24)			MCWB	47.3	47.2	48.5	51.2	55.4	60.0	62.5	61.7	57.8	55.5	51.5	49.5	(24)
(25)		10%	DB	48.1	48.0	50.1	54.0	60.7	64.9	68.2	66.5	61.4	57.4	52.1	49.7	(25)
(26)			MCWB	45.7	45.2	47.1	48.8	53.7	58.5	61.1	59.9	56.4	54.2	50.1	47.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	51.5	52.6	57.8	63.0	66.2	67.0	66.6	63.8	60.5	55.4	53.6	(27)
(28)			MCDWB	53.5	54.2	56.7	66.1	71.7	75.5	75.6	78.3	72.1	63.8	57.1	56.0	(28)
(29)		2%	WB	49.3	49.3	50.4	54.7	59.6	63.4	65.0	64.6	61.0	57.9	53.4	52.0	(29)
(30)			MCDWB	51.3	52.0	53.0	60.8	67.4	71.2	72.9	73.0	65.9	60.9	55.1	54.2	(30)
(31)		5%	WB	47.6	47.6	48.9	51.9	56.7	61.1	63.4	62.9	58.9	56.2	52.0	49.7	(31)
(32)			MCDWB	49.6	50.0	51.4	56.0	63.0	67.1	69.7	69.0	62.5	58.5	53.6	51.4	(32)
(33)		10%	WB	46.0	45.7	47.5	49.8	54.5	58.9	61.8	61.1	57.4	54.8	50.4	48.0	(33)
(34)			MCDWB	48.0	47.9	49.9	53.3	59.8	64.2	67.1	65.4	60.5	57.1	52.0	49.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	7.7	8.6	10.6	12.0	11.3	10.8	10.6	10.2	8.9	7.5	6.6	(35)
(36)			MCDBR	9.1	9.5	10.1	15.4	18.5	18.1	17.2	17.8	13.7	11.1	7.9	8.3	(36)
(37)			MCWBR	8.7	7.8	7.5	9.2	9.9	10.1	9.0	8.5	8.1	7.9	6.8	8.0	(37)
(38)		5% WB	MCDBR	8.7	9.0	9.6	14.1	17.4	15.8	14.6	14.7	13.0	9.7	8.1	7.9	(38)
(39)			MCWBR	8.6	8.0	7.7	9.1	10.1	9.9	8.4	8.2	8.7	7.7	7.5	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.339	0.368	0.390	0.394	0.391	0.392	0.378	0.365	0.348	0.325	0.314	(40)	
(41)		taud	2.320	2.382	2.316	2.275	2.309	2.343	2.346	2.400	2.437	2.443	2.386	2.293	(41)	
(42)		Ebn at Noon	200	240	256	266	271	272	270	267	256	234	201	179	(42)	
(43)		Edh at Noon	19	24	32	37	38	37	36	33	28	23	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	197	415	783	1213	1548	1615	1532	1257	893	494	242	149	(44)	
(45)		RadStd	20	45	62	107	113	146	149	81	66	42	25	15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page